

FACULTY OF ENGINEERING AND TECHNOLOGY

**ASSIGNMENT FOR THE
BSC (HONS) IS (BUSINESS ANALYTICS); YEAR 3
BSC (HONS) IS (DATA ANALYTICS); YEAR 3**

**ACADEMIC SESSION APRIL 2026
IST3244: ADVANCED BUSINESS ANALYTICS**

DEADLINE: 19th Jul 2026 Via eLearn

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INSTRUCTIONS TO CANDIDATES

- This group assignment will contribute 20% to your final grade.

Learning Outcome:

LO2: Determine which analytical techniques to apply given the nature of the problem and available data.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- Coursework submitted after the deadline but within 1 week will be accepted for a maximum mark of 40%.
- Work handed in following the extension of 1 week after the original deadline will be regarded as a non-submission and marked zero. Plagiarism and AI generated content should not be more than 20%.

Lecturer's Remark (Use additional sheet if required)

I, Keertana A/P Subramaniam, Ayu Wen Li, Siow Qi Yung, Soong Shuki, Wong Hui San (Name) 23109614, 22017867, 22053037, 23105059, 22034540 std. ID received the assignment and read the comments keertana, wenli, qiyoung, shuki, huisan (18/07/2026) (Signature/date)

Academic Honesty Acknowledgement

"I Keertana A/P Subramaniam, Ayu Wen Li, Siow Qi Yung, Soong Shuki, Wong Hui San (student name) verify that this paper contains entirely our own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties (*refer to page 16, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme*) for any kind of copying or collaboration on any assignment."

keertana, qiyoung, huisan, shuki, wenli (18/07/2026)

..... (Student's signature / Date)

1.0 Introduction

Business analytics helps organizations by playing a crucial role to make decisions that are data-driven by identifying patterns, trends, and anomalies within a dataset. In this assignment, a real-world transaction dataset by a Brazilian E-Commerce named Olist was utilized. It contains more than 100,000 orders that were placed from 2016 to 2018 across many online platforms in Brazil. This dataset was chosen because it contains comprehensive information on customer orders, products, shipping costs and product sales. This makes it suitable for analysing and data visualization. The main objective of this business analysis is to identify sales trends over time, evaluate performance of different product categories and regions, observe anomalies within the data, and to generate business insights so it can be used to support strategic decision-making and improve the performance of the business.

2.0 Dataset description

A total of 5 datasets is used in this study, olist_orders_dataset, olist_order_items_dataset, olist_products_dataset, product_category_name_translation, and olist_customers_dataset.

olist_orders_dataset

It acts as the primary operational backbone, providing order lifecycle timestamps to measure delivery performance, logistics efficiency, fulfilment delays, and supply chain operations. Table 1 shows the detail of data.

Variable	Data Type	Measurement	Key Variable	Description
order_id	Categorical	Nominal	Primary Key, Unique	Order unique identifier
customer_id	Categorical	Nominal	Foreign Key	Unique customer identifier for each order.
order_status	Categorical	Nominal	Not null	Reference to the order status
order_purchase_timestamp	Timestamp	Interval	Not null	Shows the purchase timestamp.
order_approved_at	Timestamp	Interval	Not null	Shows the payment approval timestamp.
order_delivered_carrier_date	Timestamp	Interval	Not Null	Order handover timestamp to logistics partner.
order_delivered_customer_date	Timestamp	Interval	Not Null	Order handover timestamp to customer.
order_estimated_delivery_date	Timestamp	Interval	Not Null	Estimated delivery date at purchase.

olist_order_items_dataset

It plays as the core fact table, containing the detailed products within each order. It provides the key financial data to analyze product margins and the financial impact of logistics

Variable	Data Type	Measurement	Key Variable	Description
Order_id	Categorical	Nominal	Composite Primary Key, Foreign Key	order unique identifier
order_item_id	Categorical	Nominal	Composite Primary Key	Sequential item number within an order.
product_id	Categorical	Nominal	Foreign Key	product unique identifier
seller_id	Categorical	Nominal	Foreign Key	seller unique identifier

price	Numerical	Interval	-	item price
freight_value	Numerical	Interval	-	Freight cost allocated per item.

olist_products_dataset

It serves as a dimension table that stores product attributes to support segmentation, performance analysis, evaluate how physical characteristics affect shipping costs and delivery performance.

Variable	Data Type	Measurement	Key Variable	Description
product_id	Categorical	Nominal	Primary Key	unique product identifier
product_category_name	Categorical	Nominal	-	Root product category in Portuguese.
Product_name_length	Numerical	Interval	-	Number of characters in the product name.
product_description_length	Numerical	Interval		Number of characters in the product description.
product_photos_qty	Numerical	Ordinal		Number of product photos published.
product_weight_g	Numerical	Interval		Product weight in grams.
product_length_cm	Numerical	Interval		Product length in cm
product_height_cm	Numerical	Interval		Product height in cm
product_width_cm	Numerical	Interval		Product width in cm

product_category_name_translation

It function as a bridge between raw database entries and stakeholders. Through translating raw data from Portuguese to English, ensures clearer presentation in business reports.

Variable	Data Type	Measurement	Key Variable	Description
product_category_name	Categorical	Nominal	-	category name in Portuguese
product_category_name_english	Categorical	Nominal	-	category name in English

olist_customers_dataset

It operates as a dimension table contains customer demographic and geographic information to supports regional segmentation to identify key markets, and potential supply chain improvements.

Variable	Data Type	Measurement	Key Variable	Description
customer_id	Categorical	Nominal	Primary Key, Unique	Unique customer identifier for each order.
customer_unique_id	Categorical	Nominal	Foreign Key	Unique identifier of a customer.
customer_zip_code_prefix	Spatial	Nominal		First five digits of the customer ZIP code.
customer_city	Spatial	Nominal		Customer location city name
customer_state	Spatial	Nominal		Customer location State.

3.0 Data Pre-Processing

The Olist dataset was pre-processed in Power BI to improve data quality and consistency. Five tables were imported which were olist_orders_dataset, olist_order_items_dataset, olist_products_dataset, product_category_name_translation, and olist_customers_dataset. When importing, the ‘Use First

Row as Headers' function was applied to the product_category_name_translation table so that the column names were identified.

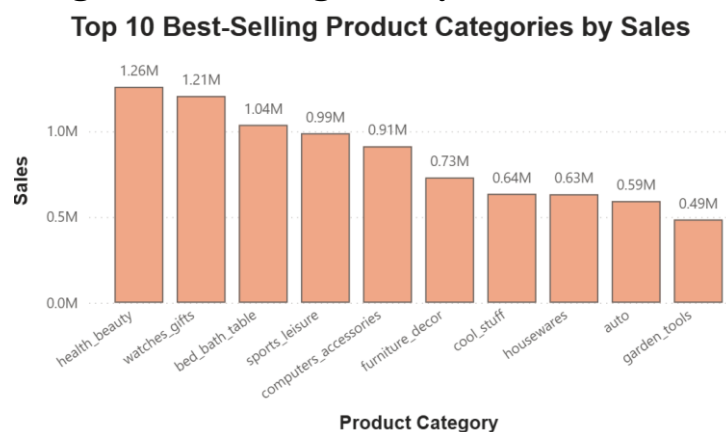
Next, the olist_products_dataset was merged with the product_category_name_translation table using the product_category_name field so that the product category name that were in Portuguese were translated to English. Then the original product_category_name that was in Portuguese was removed to improve readability. Later, the data type for order_purchase_timestamp, order_delivered_customer_date, order_estimated_delivery_date, and shipping_limit_date were changed to Date to support time-based analysis.

Missing values were also reviewed. However, since the records with missing values corresponded to valid order statuses, such as Invoiced, Cancelled, and among others, these records were retained as they were not data entry errors. Similarly, missing values in the product_category_name_english were replaced with 'Uncategorized' so that all sales records were included in the analysis.

Additionally, a new column named delivery_days was created by calculating the difference between the date of purchase and the actual delivery date. Another column named total_order_value was also added which contains the sum of price and freight_value fields to represent the total transaction value. Finally, the relationships between the tables were established using the customer_id, order_id, and product_id fields so that a relational data model was created for subsequent visualization and analysis.

4.0 Visualizations

4.1 Top 10 Best-Selling Product Categories by Sales

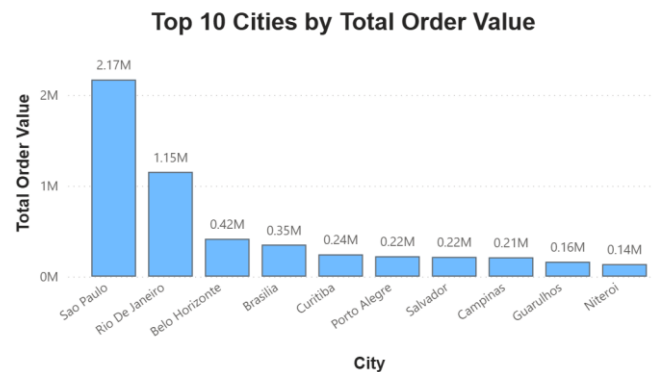


The bar chart shows the top 10 best-selling product categories by sales. This visualization aims to identify the product categories that generate the highest product sales. This allows businesses to understand customer purchasing preferences and prioritize strategies for inventory planning, marketing campaigns and resource allocation. Additionally, these insights can improve product assortment decisions and allocate resources effectively to maximize revenue growth while maintaining customer satisfaction.

The bar chart shows that health & beauty generates the highest revenue with 1.26 million, followed by watches & gifts with 1.21 million and bed, bath & table with 1.04 million. These are the top three product categories with the highest revenue. They are followed by sports & leisure, computer accessories, furniture & home décor, lifestyle products, household items, automotive products and garden tools. This shows that product categories including health beauty, watches gift and bed bath tables generate the highest revenue. These categories have strong sales performance and contribute

significantly to overall business revenue. This also shows that customers spend more on products related to personal care, lifestyle and household needs. Therefore, business managers and marketing teams should focus more on these high-revenue categories by ensuring enough stock is available, creating targeted promotions and offering suitable deals to attract more customers and increase sales.

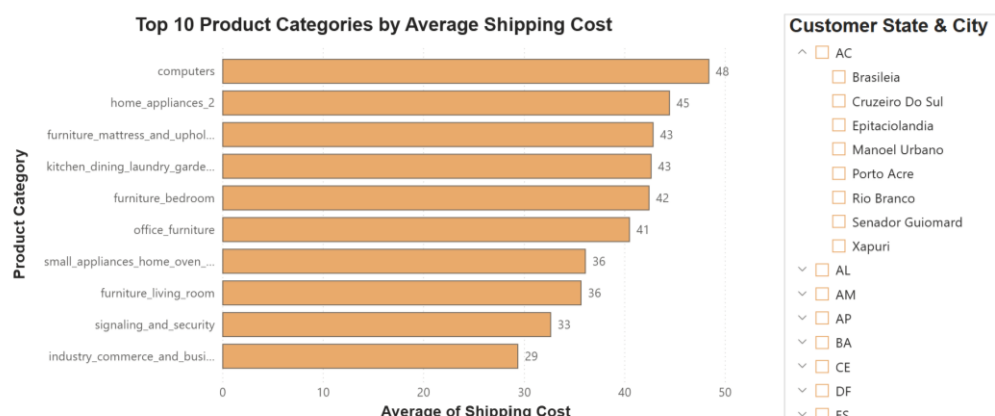
4.2 Top 10 Cities by Total Order Value



The bar chart shows the top 10 cities by total order value. This visualization aims to identify the cities that generate the highest order value. This will help businesses to identify important markets with stronger customer demand, allowing them to develop business strategies and make better plans for future growth.

The bar chart shows that Sao Paulo contributes the highest total customer payment with 2.17 million, followed by Rio De Janeiro with 1.15 million and Belo Horizonte with 0.42 million. They are followed by Brasilia, Curitiba, Porto Alegre, Salvador, Campinas, Guarulhos and Niteroi. This shows that customers from Sao Paulo, Rio De Janeiro and Belo Horizonte in Brazil contribute significantly to the overall business revenue. These cities are important markets with higher purchasing activity. Therefore, business managers and marketing teams should focus on these cities by improving customer engagement, creating marketing strategies and ensuring efficient delivery services to serve customers in these areas better.

4.3 Top 10 Product Categories by Average Shipping Cost



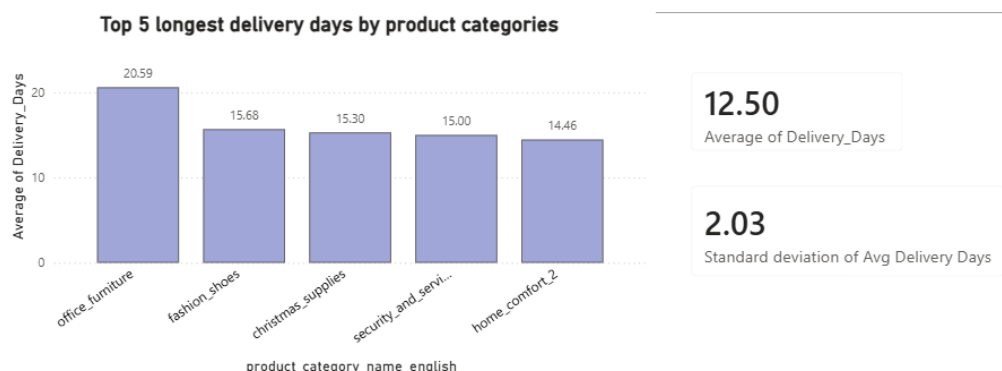
This data visualization uses a bar chart to analyze the Top 10 Product Categories by Average Shipping Cost. The chart compares product categories based on the average freight_value, representing the shipping cost associated with each order. Categories with higher average shipping costs may require more expensive transportation due larger size, heavier weight or special handling requirements.

An Average Item Price tooltip is added to allow users to compare shipping costs with item prices, providing context on whether higher shipping expenses are reasonable. Additionally, Customer State and City slicers enable users to explore geographical differences in shipping expenses and identify locations with higher delivery costs.

The bar chart shows that Computers have the highest average shipping cost of \$48 across all customer states and cities. This may be due to these products often requiring larger packaging, higher product weight and additional protection for fragile components during transportation.

From a business perspective, this visualization helps identify product categories and areas with higher shipping costs, allowing businesses to understand their logistics challenges. These insights can help companies review shipping strategies, improve delivery processes, optimize warehouse locations and make better pricing decisions. By comparing shipping costs across categories and locations, businesses can identify factors affecting delivery expenses and focus on areas where costs can be reduced.

4.4 Average Delivery Date used by each Product categories

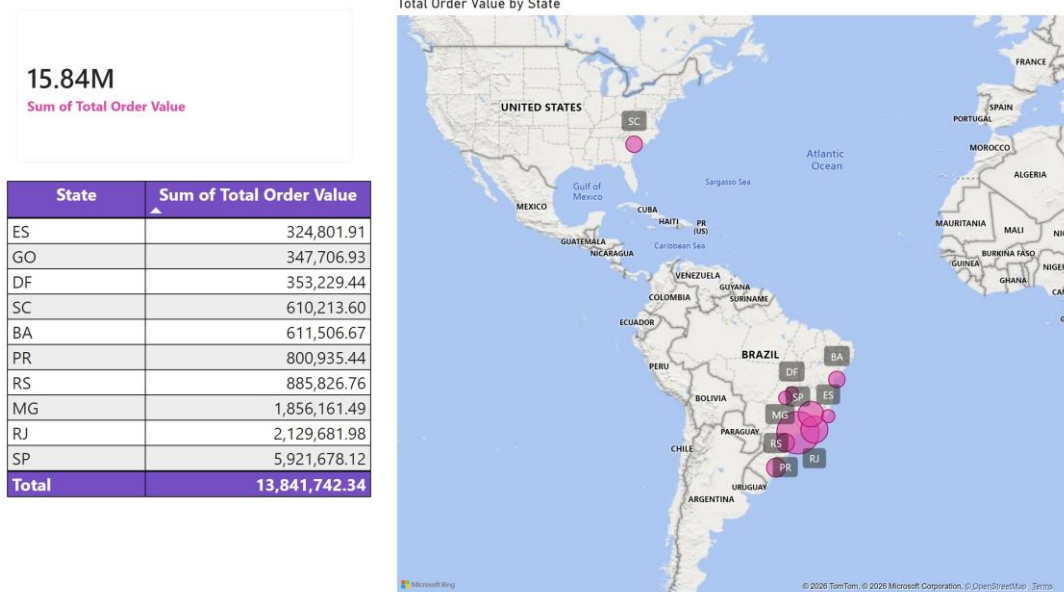


The bar chart represents the overall average delivery days, the standard deviation calculated from each product category average delivery days, and the top 5 Longest Delivery Days by Product Categories. This analysis identifies product categories with poor delivery performance, indicated by delivery days that exceed the benchmark.

The average delivery time across all orders is 12.5 days. As this analysis classifies product categories into either good or bad delivery performance (50% probability each), 12.5 days is used as the initial benchmark. Categories that have an average delivery date under 12.5 days will be considered as good performance.

However, average delivery days is not a sufficient benchmark, as a category with slightly above doesn't necessarily indicate poor performance. Thus, the standard deviation, 2.03 days, across category averages is calculated, measuring the difference typical category average deviates from the overall mean. Combining both measures, the benchmark for poor delivery performance is 14.53 days (12.5 + 2.03 days). Through comparing the benchmark with the Top 5 Longest Delivery Days by Product Categories, 4 group of product categories show a bad performance in delivery which is office_furniture (20.59days), fashion_shoes (15.68days), Christmas_supplies (15.30days) and security and services (15days).

4.5 Top 10 States by Total Order Value



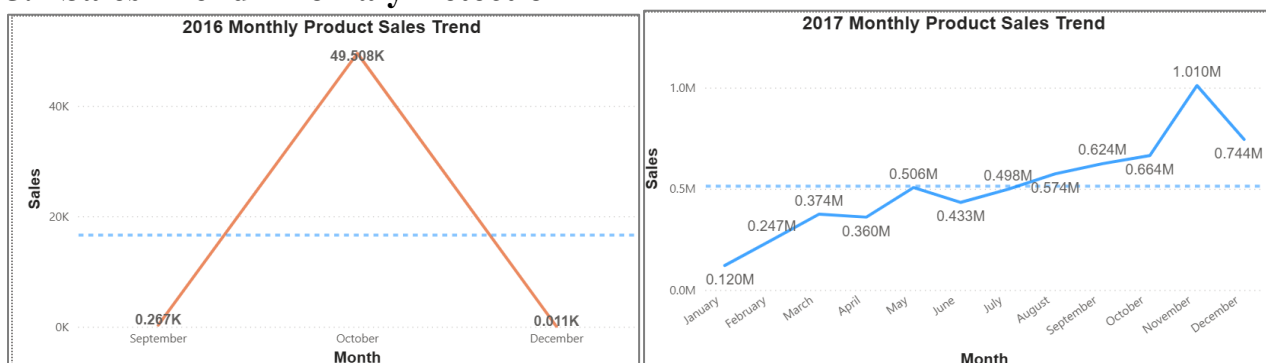
The chart above features a table and a map that depicts the top 10 states with the highest total order value with the objective of identifying the strongest regional markets across the states of Brazil. Through this analysis, businesses will be able to understand the regions with the greatest customer demand and business potential by comparing sales performance across the states, allowing more effective regional marketing and sales strategies.

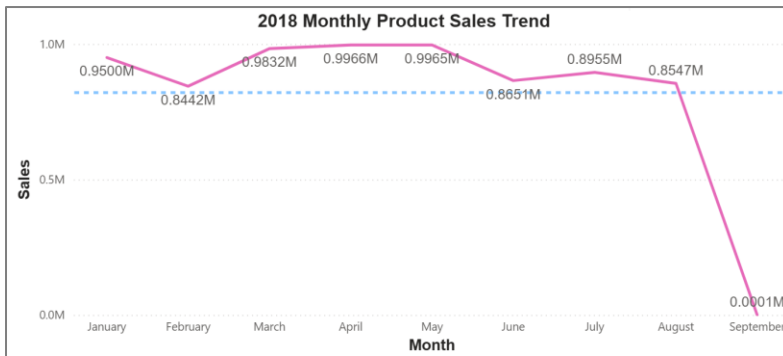
The results show that Sao Paulo (SP) significantly outperformed the other states, recording \$5921678.12 in total order value. Rio de Janeiro (RJ) comes in second with total order value of \$1856161.49, followed by Rio Grande do Sul (RS) and Parana (PR) with \$885826.76 and 800935.44 total order value respectively. This is followed by Bahia (BA), Santa Catarina (SC), Distrito Federal (DF), Goias (GO) and Espirito Santo (ES).

This highlights the high concentration of Brazil's southeastern region for total order value, indicating the primary customer market, high stock turnover and order value driver for Olist. States with the highest order value should be priorities in terms of promotional initiatives, customer engagement and inventory allocation strategies. In contrast, the low performing states suggest market expansion potential through localised marketing campaigns.

5.0 Anomaly detection and analysis

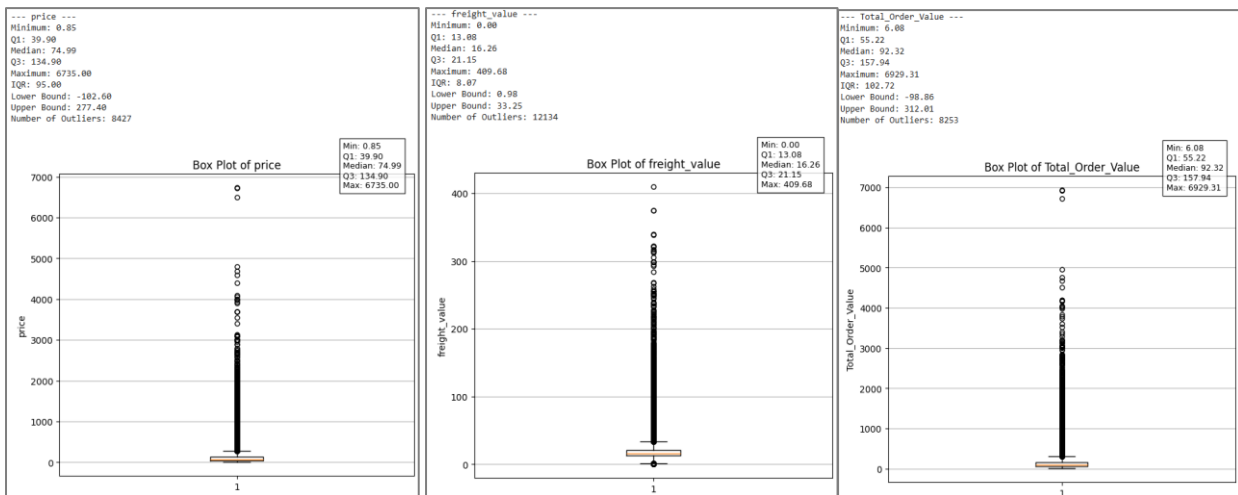
5.1 Sales Trend Anomaly Detection





The line charts show some significant increases and decreases throughout the period. In 2016, from September to October, sales increased significantly because September data may not fully cover the entire month and represent the actual monthly sales performance. The increase may also be influenced by higher customer demand or increased order activity in October. Also, from October to December 2016, sales dropped sharply which may also be affected by incomplete data records or caused by lower order activity. In 2017, from October to November, sales increased significantly, possibly due to higher customer demand, seasonal shopping activities or promotions. However, from November to December, sales declined significantly which may be caused by lower customer demand after the November peak, fewer promotions or changes in customer purchasing behaviour. In 2018, from August to September, sales decreased may be influenced by low order activity during that period.

5.2 Boxplot Analysis



The boxplot for price has a median value of 74.99 while the interquartile range (IQR) is 95.00, indicating that most customers' orders are affordable products. However, there are numerous high, highlighting that the few-price items increased the average product price. The second boxplot for freight value has a median value of 16.26 and small IQR which reveals that the majority of the shipping costs are inexpensive with little variation among the orders. Freight value also has the highest count of outliers, which suggests a significant number of orders having high shipping charges.

The third boxplot for total order value combines both price and freight value. As freight costs are usually small relative to item price, the median and maximum values are similar to the price boxplot. Likewise, the outlier count is comparable to the price. The high count of outliers reveals that the spread of total income among orders is disproportionate with only a small portion of orders generating most of the order value. This is due to the low, affordable pricings of most of the product items and few highly expensive items, making the ecommerce data to be heavily skewed to the right.

6.0 Conclusion

The visualizations show that Health & Beauty, Watches & Gifts, and Bed, Bath & Table are the highest revenue-generating product categories, indicating strong customer demand for personal care, lifestyle and household products. São Paulo, followed by Rio de Janeiro and Belo Horizonte, contributes the highest customer payments, making these cities the company's most important markets. From a logistics perspective, the Computers category has the highest average shipping cost, while Office Furniture, Fashion Shoes, Christmas Supplies, and Security & Services exceed the delivery benchmark of 14.53 days, indicating weaker delivery performance.

Based on these insights, the company should prioritize inventory allocation and marketing investment for the top-performing product categories in high-spending cities. Therefore, strategies such as maintaining higher inventory levels, targeted promotions, and sales forecasting can be implemented to increase sales while reducing stockout risks during peak demand periods.

To improve profitability, the company should optimize logistics for categories with high shipping costs, particularly Computers. This can be achieved by positioning inventory closer to high-demand cities and negotiating better shipping rates with logistics providers. Additionally, the company can use more suitable packaging to reduce dimensional weight and lower transportation costs without affecting service quality.

For product categories with long delivery times, the company should review its warehouse operations, delivery services, and order handling processes to identify the causes of delays. The company can regularly evaluate delivery partners to improve delivery speed. Business dashboards can also be used to track performance and support better decision-making based on customer demand.

7.0 Group Contribution Table

Member	Contribution
Keertana A/P Subramaniam (23109614)	Proposed the dataset, prepared the Introduction, performed Data Cleaning and Preprocessing section, designed the overall Power BI dashboard layout, reviewed the overall report, and assisted in compiling and formatting the final report.
Ayu Wen Li (22017867)	Proposed the dataset, developed Data Visualization 4.5 and its description, contributed to Anomaly Detection & Analysis, reviewed the report.
Soong Shuki (23105059)	Proposed the dataset, prepared the Dataset Description, developed Data Visualization 4.4 and its description, and reviewed the report.
Siow Qi Yung (22053037)	Proposed the dataset, developed Data Visualization 4.1 & 4.2, and its description, contributed to Anomaly Detection & Analysis, and reviewed the report.
Wong Hui San (22034540)	Proposed the dataset, developed Data Visualization 4.3 and its description, prepared the Business Insights & Proposal, and reviewed the report.

References

Olist, & Sionek, A. (2018). *Brazilian E-Commerce Public Dataset by Olist* [Dataset]. Kaggle. <https://doi.org/10.34740/KAGGLE/DSV/195341>

Appendix

Power BI Dashboard:

<https://app.powerbi.com/view?r=eyJrIjoiM2Q0MjVhNDgtMzU2NC00NzMyLWI2Y2MtNjU5MDBkNTQyZDczliwidCI6IjUzN2MyYmUxLWZjZDQtNDVhOS04M2IzLTUyNTNlYWwNTA3MCIslmMiOiJlZWfQ%3D%3D&pageName=d504e1ad2e5be1aad1dc>

Visualization for Box Plot (Anomaly Detection): https://colab.research.google.com/drive/1B-QScTEBQ_8Fa1yQ3oTRrlcq7EMH1jib?usp=sharing